

Jerry Hsu

Profile

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Education

University of Utah
Doctor of Philosophy (PhD)

Salt Lake City, Utah
Aug 2020 - May 2025

Research topics: Physically Based Animation, Physics Simulations
Advisor: Prof. Cem Yuksel

Purdue University
BS. Computer Science
BS. Mathematics

West Lafayette, Indiana
Aug 2017 – May 2020
Aug 2017 – May 2020

GPA 3.81

Research Publications

Divide and Truncate: A Penetration and Inversion Free Framework for Coupled Multi-physics System

ACM SIGGRAPH 2026

Anka Chen, **Jerry Hsu**, Youssef Ayman, Miles Macklin

Implicit Position-Based Fluids

ACM SIGGRAPH ASIA 2025

Elie Diaz, **Jerry Hsu**, Eisen Montalvo-Ruiz, Chris Giles, Cem Yuksel

Neighbor-Aware Data-Driven Relaxation of Stitch Meshes

ACM SIGGRAPH ASIA 2025

Yura Hwang, Jenny Han Lin, **Jerry Hsu**, Benjamin Mastropolito, James McCann, Cem Yuksel

Stable Cosserat Rods

ACM SIGGRAPH 2025

Jerry Hsu, Tongtong Wang, Cem Yuksel, and Kui Wu

Offset Geometric Contact

ACM SIGGRAPH 2025

Anka Chen, **Jerry Hsu**, Ziheng Liu, Miles Macklin, Yin Yang, Cem Yuksel

Real-Time Physically Guided Hair Interpolation

ACM SIGGRAPH 2024

Jerry Hsu, Tongtong Wang, Zherong Pan, Xifeng Gao, Cem Yuksel, and Kui Wu

- This work introduces a novel physically-driven hair interpolation scheme that utilizes existing simulated guide hair data. This new scheme enables the efficient and artifact free interpolation of hair.

Sag-Free Initialization for Strand-Based Hybrid Hair Simulation

ACM SIGGRAPH 2023

Jerry Hsu, Tongtong Wang, Zherong Pan, Xifeng Gao, Cem Yuksel, and Kui Wu

A General Two-Stage Initialization for Sag-Free Deformable Simulations

ACM SIGGRAPH 2022

Jerry Hsu, Nghia Truong, Cem Yuksel, and Kui Wu

- A general framework to efficiently solve for sag free initializations applicable to cloth, FEM, MPM, and PBD based simulations.

Intelligent Adaptive Splitting for Vector Representations

Awaiting submission

Industry Positions

Zeromatter

Research Engineer

- **Continuing work on high fidelity physics engines**

Mountain View, California
2025-Current

Tencent Lightspeed & Quantum Studios

Research Intern

- **Real-Time Physically Guided Hair Interpolation** extends simulation quantities to interpolation for robust, efficient, and responsive hair animations.

Los Angeles, California
Summer 2023

Tencent Lightspeed & Quantum Studios

Research Intern

- Performed research, implementation, and experimentation on state-of-the-art hair simulation methods.
- Sag-free initialization of said simulation performed by the inverse solve of hair rest configurations to counter sagging due to gravity.

Los Angeles, California
Summer 2022

Adobe Research Computational Artistry Team

Research Intern

- Performed research through mathematical derivation to implementation and experimentation.
- **Intelligent adaptive splitting**: A representation independent way of optimizing over discrete parameters in vector graphics.

San Jose, California
Summer 2021

Amazon

SDE Intern

- Implemented diagnostic data processing on cloud systems for bill computation. Designed data chunking systems that reduced network transfer usage by 70% when accessed.

Seattle, Washington
Summer 2019

Other

Technical Skills

- Proficient languages: Rust, C++, CUDA, GLSL/HLSL, Mathematica
- Proficient in: GPU performance optimization, numerical optimization, PDE modeling, physics simulations

